Military Specification Sheet

TRANSMISSION, POWER, CONSTANT SPEED, 60/25 KW

The complete requirements for procuring the transmission described herein consist of this document and Military Specification MIL-T-7101A of 15 Sep 1951, hereinafter noted as MIL-T-7101.

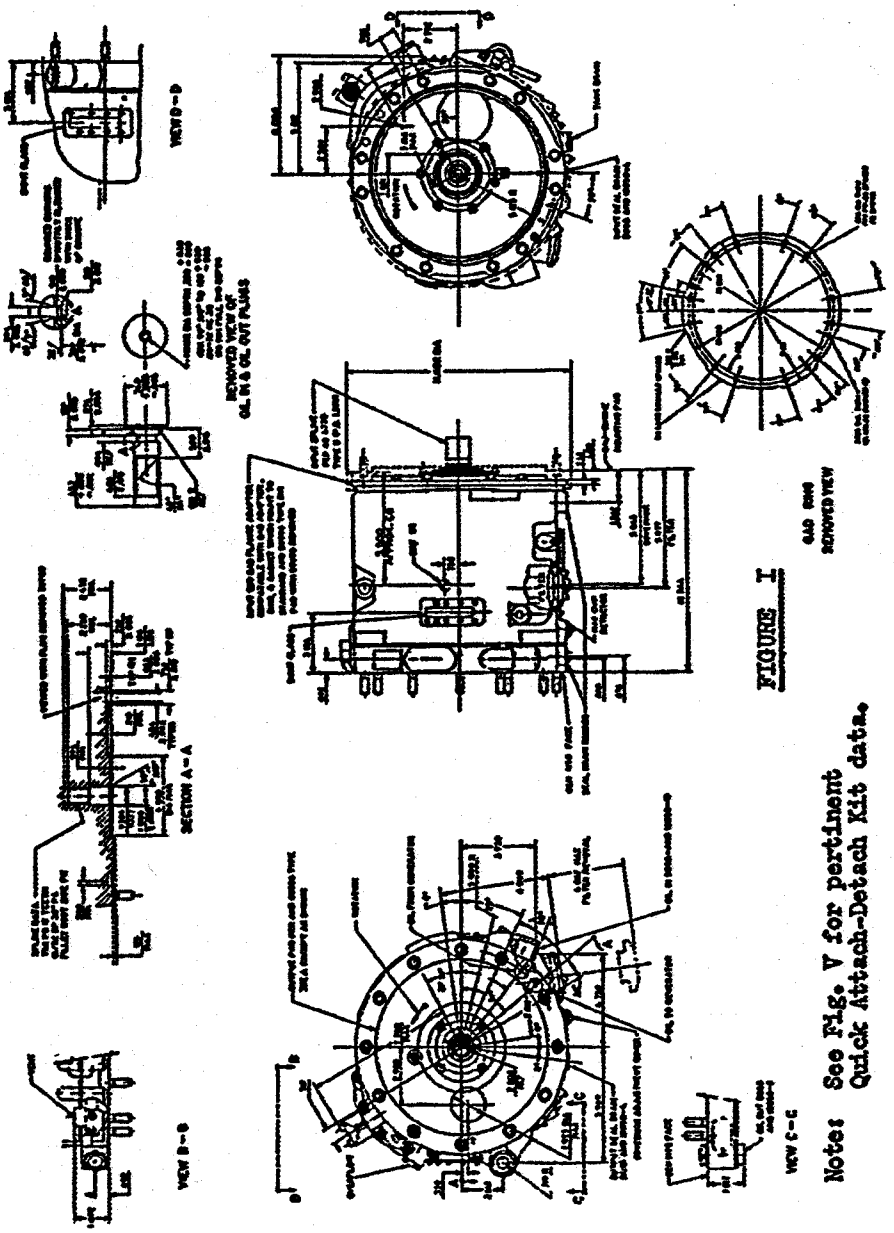


FIGURE I

Notes: See Fig. V for pertinent Quick Attach-Detach Kit data.

FSC 6115

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GENERAL REQUIREMENTS:

Dimensions and configuration: FIG. I
Dimension tolerances: fractions ±1/16
decimals ±.06
angles ±1°
Rating data: Table 1
Heat rejection: FIG. II
Color: Federal Standard 595 Color 17875
Part Number: M7101/1-60

CHANGES TO SPECIFICATION REQUIREMENTS:
The requirements of MIL-T-7101 apply with the following exceptions to existing paragraphs:
3.1.2- Change Drawing AND 10398 to Military Standard MS 33586.
3.1.3- Change ANA Bulletin No. 143 to MIL-STD-143.
3.1.3.1- Delete AN and add MS throughout.
3.2- Delete and Add: The transmission shall operate in aircraft for 1000 hours at a 60 KW rating or for 2500 hours at a 25 KW rating under the following conditions or natural combination of conditions, except that operation at zero or negative gravitational units (g) for periods exceeding 30 consecutive seconds is not required.
3.2.1- Delete and Add: Constant or varying ambient pressure ranging from 22.7 to 1.68 psia. Maximum pressure variation of 5 psi per second.
3.2.2- Delete and Add: Ambient temperature from -54° C to 149° C.
3.2.4- Delete and Add: Sand and dust as encountered in desert areas.
3.3.3.1- Delete "of a non-inflammable type".
3.3.6- Delete and Add: The transmission shall introduce no harmful torsional vibrations between its input and the generator, nor shall it increase the amplitude of any torsional vibration applied to the input end to a harmful degree.
3.3.7.2- Add: The transmission shall be also capable of withstanding, without damage, reverse rotation of the input shaft of up to 100 rpm.
3.3.8- Delete and Add: There shall be no leakage of oil from the transmission to the generator. All leakage from the rotating seals shall be through drain ports provided on the transmission. Total leakage shall not exceed 2 cc per hour per rotating seal under any operating or static condition. External case leakage shall not exceed 4 drops per hour.
3.3.10- Delete and Add: Rotation shall be as indicated on Figure I of the Specification Sheet.
3.3.11- Change "one hour" to "100 hours".
3.3.11.1- Delete and Add: The transmission shall incorporate an automatic decoupler completely contained within the transmission envelope to protect the aircraft against a ruptured transmission or excessive transmission temperatures. The transmission shall automatically decouple from its prime mover prior to any external mechanical damage to the transmission/generator package, the average case temperature exceeding 204°C as a result of transmission failure, or the internal oil temperature reaching 204° C. Force required to accomplish a disconnect shall be derived from the prime mover and not from a separate energy source. The decoupler shall not be resettable until disassembly of the transmission. Subsequent to decoupler operation, the transmission shall withstand four hours of operation at maximum input shaft speeds. Operation of the decoupler during any qualification test other than that provided by 4.6.3 shall be cause for rejection.
3.3.12- Delete and Add: Speed control adjustments shall not be required during installation or normal service life of the transmission. If external adjustment provisions are provided, means shall also be provided to preclude accidental adjustment.
3.3.13- Delete and Add: Upon starting, the transmission output speed shall be 8000±80 rpm within 5 seconds after the input speed has reached 3700 rpm at ambient temperatures above -40°C. At temperatures between -40°C and -54°C output speed shall be 8000 ±400 rpm within 5 seconds and 8000±80 rpm within 1 minute after input speed has reached 3700 rpm.
3.3.15- Add: "at a rating of 60 KW, or 2500 hours at a rating of 25 KW."
3.3.16- Add: The transmission shall require no energization from any external source except the input drive shaft during starting. The transmission shall be self-priming and capable of making starts without damage with the oil supply line initially filled with air.
3.10- Add: "in accordance with MIL-STD-130".
3.10.1- Delete and Add: Each transmission nameplate shall be in accordance with MIL-P-

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6906 and shall contain the following information:

Transmission, Constant Speed, A.C.Generator
Manufacturer's Part Number
Manufacturer's Serial Number
Federal Stock Number
Military Part Number M7101/1-60
Contract Number
Date of Manufacture
Manufacturer's Name or Trademark
US

Decals shall not be used as permanent nameplates."

4.5.2- Delete: "in revolutions per minute."

4.5.3- Delete and Add:Total oil leakage from the transmission shall be collected and measured during all operating and static conditions while under test. The leakage shall not exceed the limits of 3.3.8.

4.6.2.1- Delete and Add: The high temperature test shall be in accordance with MIL-E-5272 Procedure II modified as follows: The transmission shall be subjected to an ambient temperature of 149°C for 12 hours before operating. At the end of this period and while still subjected to this temperature, the transmission shall be operated at rated load and average rated input speed for 5 hours. Inlet oil temperature to the reservoir shall be maintained at 121±5°C. The transmission shall complete this test without failure or subsequent impairment of performance.

4.6.2.2- Delete and Add: Low temperature tests shall be in accordance with MIL-E-5272 Procedure II modified as follows:The transmission shall be subjected to an ambient temperature of -54°C for at least 72 hours before operating. At the end of this period and while still subjected to this temperature the transmission shall be started and accelerated to minimum input speed at a minimum rate of 200 rpm per second. The transmission shall be operated without load for 5 minutes at minimum input speed while the oil temperature is raised to 37.8°C. Transmission output speed regulation shall be within the limits of 3.8.3.1 at the end of the 5 minute warm-up period.

4.6.3- Delete and Add: The transmission shall be tested with input speed, load, and temperature varied in accordance with Table 2. The transmission shall operate in accordance with Table 2 for 1000 hours. Oil shall be replaced each 250 hours. The filter shall be serviced each 100 hours. The decoupler shall be operated 100 times during this test with decoupler operation divided equally with respect to specified load and speed conditions. For the first 95 decoupler operations, an electrically actuated device may replace the thermally actuated device specified in 3.3.11.1.

4.6.11- Delete "no load" and add "4 horsepower" throughout.

4.6.13.2- Change "4%" to "5%".

4.6.15- Add: "Interference limits shall be in accordance with MIL-I-6181."

4.6.17- Delete and Add: The transmission shall be shock tested in accordance with MIL-E-5272 Procedure V. The transmission shall be installed on a test fixture which simulates the engine accessory pad. An actual or dummy generator with larger overhung moment than that specified for use with the transmission may be used for this test.

4.6.18- Delete and Add: The transmission shall be installed on test fixtures which simulate the aircraft installation and shall be tested in accordance with MIL-E-5272 Procedure XII, except that the tests shall be conducted at room temperatures and in accordance with Figure III of the Specification Sheet.

4.6.19- Add: "The acoustical noise level shall not be raised more than 5 db under any condition described in the Specification Sheet."

4.6.21- Delete and Add: Operating conditions imposed by the aircraft installation and aircraft flight manuevers shall be simulated for laboratory tests. The transmission shall be installed and tested on a stand which can be rotated up to 180° about each of the two perpendicular axes. The transmission shall be operated in accordance with 3.2.7.1 and 3.2.7.2 while carrying 100% rated load.

4.6.22- Delete and Add: The transmission shall be subjected to a humidity test in accordance with MIL-E-5272 Procedure I.

4.6.23- Delete and Add: The transmission shall be subjected to a salt spray test in

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accordance with MIL-E-5272 Procedure I.
4.6.24- Delete and Add: The transmission shall be subjected to a sand and dust test in accordance with MIL-E-5272 Procedure I.
4.6.25- Delete and Add: The transmission shall be subjected to a fungus test in accordance with MIL-E-5272 Procedure I.
TABLE I (MIL-T-7101): Change as follows:
Part B Test- Change "9" to "g".
Note (b)- Add: "with inlet oil temperature at 121±3°C."
Note (d)(1)- Change "no load" to "¼ horsepower".
Note (d)(3)- Delete "input speeds below" in the first line.
Note (h)- Delete "input speeds below" in the second line.
Note (h)- Change "no load" to "¼ horsepower".
Note (i)- Change "no load" to "¼ horsepower".

ADDITIONAL REQUIREMENTS:
The following requirements are in addition to those of MIL-T-7101 as modified above:

3.2.7.1 Positions - The transmission shall perform in accordance with the requirements of this specification when placed in the following positions under the conditions specified and shall continuously perform in accordance with the requirements of this specification during change of position and/or gravitational conditions specified. Changes in position may occur rapidly and repeatedly during flight with no definite time lag between conditions.
(1) ANY POSITION under the gravitational conditions of 3.2.7.2 except sustained zero or negative g conditions-CONTINUOUSLY.
(2) ANY POSITION under sustained zero or negative g conditions-30 SECONDS.
3.2.7.2 Flight Manuever Forces - The transmission, including mounting provisions, shall safely withstand 1.15 times the loads due to the following aircraft limit load factors (g), angular accelerations (α), and velocities (w) at the aircraft center of gravity, and shall withstand 1.50 times these forces without failure. The aircraft center of gravity shall be 18.5 inches above, 105.7 inches forward, and 6.1 inches to the left of the center of the mounting flange at the transmission. The transmission shall operate satisfactorily in accordance with the requirements of the Specification Sheet at any point, including zero g within an envelope formed by the following limits applied to the aircraft center of gravity: (See Figure IV for sign convention)

(a) Flight (groups of simultaneous conditions)
(1) Symmetrical: $g_z = +7.0$, $g_x = \pm 2.5$ $\alpha_y = 0$ to ± 6.0 radians per second per second, $w_y = 0$ to ± 1.5 radians per second.

$g_z = -3.0$, $g_x = \pm 2.5$, $\alpha_y = 0$ to ± 6.0 radians per second per second, $w_y = 0$ to ± 1.0 radians per second.

(2) Unsymmetrical: $g_z = +5.6$, $g_x = \pm 2.0$, $g_y = \pm 2.0$, $\alpha_y = 0$, $\alpha_x = \pm 15$, $\alpha_z = \pm 5$ radians per second per second.

$g_z = +1.0$, $g_x = \pm 2.0$, $g_y = \pm 3.0$, $\alpha_x = \pm 20$, $\alpha_z = \pm 7$ radians per second per second.

(b) Ground
(1) Catapult: $g_x = -6.1$, $g_z = +1.0$, $g_y = \pm 2.0$, $\alpha_y = +1.0$ radians per second per second.
(2) Arrested Landings: $g_x = +5.3$, $g_y = \pm 1.0$, $g_z = +9.5$, $\alpha_y = \pm 12.0$, $\alpha_x = \pm 4.0$ $\alpha_z = \pm 5.0$ radians per second per second.
(3) Field Landings: $g_x = +2.0$, $g_z = +6.0$, $g_y = \pm 1.0$, $\alpha_x = +10.0$, $\alpha_y = \pm 3.0$ radians per second per second.

3.3.2.1 Modification - The transmission shall be capable of being modified to operate with oil-cooled generators without disassembly. It shall be required only to remove two oil port plugs and to install generator bypass valve parts. The bypass valve parts shall be installed from the

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exterior of the transmission and shall be contained within the envelope shown on Figure I of the Specification Sheet. Removal of the two oil port plugs shall be from the external surface of the transmission.

3.2.8 Mechanical Shock - The transmission shall operate satisfactorily during and after encountering mechanical shock loads equivalent to 18 impact shocks of 15g for 11±1 milliseconds along the three mutually perpendicular axes and two impact shocks in each direction of 30g acceleration applied about each of the three major axes. Each shock impulse shall have a duration of 11±1 milliseconds. Bending and distortion is permitted. There shall be no failure of the attachment joints, and all parts of the transmission shall remain in place.

3.2.9 Vibration - The transmission shall operate satisfactorily when subjected to vibrations of 5 to 2000 cycles per second at an applied double amplitude of .012 inch or 6g, whichever is less.

3.3.20 Oil Reservoir - The transmission shall include an all-attitude oil reservoir and all required case pressurization and anti-drain valves as an integral part of the transmission assembly. One fill port and two sight gages for checking fluid quantity shall be provided. Fill ports shall also be usable as overfill indicators when the transmission is gravity-filled.

3.3.21 Oil Filters - Each transmission shall include an integral oil filter, removable within the clearance indicated on Figure I of the Specification Sheet. Filters shall be suitable for use with oil conforming to MIL-L-7808 or MIL-L-23699, shall conform to MIL-F-5504 or equivalent, and shall not require servicing more than once each 200 hours. The filter shall have a bypass setting of 75±10 psi. The design shall provide for easy removal and installation of the filter element without the use of special tools, and while the transmission is mounted on the engine

3.3.22 Magnetic Drain Plug - The transmission shall have a magnetic drain plug (chip detector) which shall include a check valve to prevent sump draining when the detector alone is removed for inspection. The detector shall provide for inspection for chips by means of an electrical continuity check without removal of the detector from the transmission.

3.3.23 Fluid Cooling -

3.3.23.1 Heat Rejection - The total heat rejection to the oil shall not exceed the limits of Figure II of the Specification Sheet.

3.3.23.2 Maximum Oil Temperature - The transmission shall meet the requirements of this specification with a continuous oil inlet temperature of 37.8°C to 121°C, and with a minimum oil inlet temperature of -54°C during starting.

3.3.23.3 Scavange Pressure - The transmission shall be capable of continuous operation with an external oil cooling circuit pressure drop of 30 psi to 100 psi, exclusive of the oil reservoir and the deaerator. Maximum oil pressure at the transmission inlet port shall not exceed 100 psig under normal operating conditions, including cold starts, and shall not exceed 170 psig under exceptional operating conditions such as with clogged deaerator nozzle.

3.3.24 Fail-Safe Provisions - The speed governing device design shall be fail-safe, in that in the event of an overspeed the governor speed control system will act to decrease the output speed, and in the event of failure of the governor drive mechanism, the transmission shall be placed in the minimum speed ratio.

3.3.25 Radio Interference - Interference limits shall be in accordance with the limits of MIL-I-6181.

3.8.1 Rating - Transmission rating shall be as specified in Table 1 of the Specification Sheet. This rating applies for all specified ambient and cooling conditions.

3.8.2 Input Speed

3.8.2.1 Range - As specified in Table 1 of the Specification Sheet.

3.8.2.2 Rate of Change - During input acceleration or deceleration rates up to 1500 rpm per second, transmission output speed shall not cause generator output

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frequency to exceed limits 2 and 3 of Fig. 3 of MIL-STD-704. Recovery to steady state frequency shall not exceed limits 2 and 3 of Fig. 3 of MIL-STD-704 after the end of the speed change.

3.8.2.3 Low Speed Capability - The transmission shall have the capability of rotating continuously at no load at any input speed from 0 rpm to 3700 rpm without damage, and shall be capable of carrying rated load for at least 5 minutes at an input speed corresponding to an output speed of 7300 rpm without damage.

3.8.3 Output Speed -

3.8.3.1 Regulation - Transmission output speed shall be 8000±80 rpm for all steady state conditions, and 8000±400 rpm for all transient conditions, including 150% load.

3.8.3.2 Modulation

3.8.3.2.1 Amplitude - Transmission output speed shall be such that variations of generator output frequency due to modulation during any 1 minute period shall be within a band of ± 2 cycles per second about a mean frequency. Mean frequency shall remain within the limits of 397 to 403 cycles per second.

3.8.3.2.2 Rate - Transmission output speed shall be such that rate of generator output frequency change due to modulation shall not be greater than 25 cycles per second per second.

3.8.3.3 Drift - Transmission output speed shall be such that variations within steady state limits of generator output frequency due to drift shall not be greater than ±5 cycles per second. Rate of frequency drift shall not exceed 15 cycles per second per minute.

3.8.4 Shock Loads -

3.8.4.1 Constant Input Speed - Transmission output shall be such that generator frequency deviations and recovery time shall be within limits 2 and 3 of Fig. 3 of MIL-STD-704 when any step load between 0 and 150% rated load is applied with input speed constant.

3.8.4.2 Combined Input Speed/Load Changes Transmission output shall be such that generator frequency deviations shall be within limits 2 and 3 of Fig. 3 of MIL-STD-704 for the condition of simultaneous maximum input speed acceleration or deceleration and shock load applications of up to 150% rated load. Recovery to steady state frequency condition shall not exceed these same limits after reaching the end of the speed range.

4.3.2.1 Speed Regulation - Each of the following tests shall be performed at inlet oil temperature of 121±5°C.

4.3.2.1.1 Steady State Operation - With transmission input speed of 3700 rpm, apply constant loads of 0, 50%, and 100% rated load. Repeat with input speed at 7400 rpm, and with input speed speed increasing from 4500 to 5500 rpm at a rate less than 50 rpm per second. Output speed shall remain within the limits of 3.8.3.

4.3.2.1.2 Input Speed Changes - With input speed of 3700 rpm and no load on the generator, accelerate the input speed to 7400 rpm at a rate of 900-1500 rpm per second. Repeat this test at 50% and at 100% rated load. With input speed of 7400 rpm and no load on the generator, decelerate the input speed to 3700 rpm at a rate of 900-1500 rpm per second. Repeat this test at 50% and at 100% rated load. Output speed shall remain within the limits specified in 3.8.2.2.

4.3.2.1.3 Shock Loads - With input speed of 3700 rpm, apply and remove shock loads of 0, 50%, 100%, and 150% rated load. Repeat this test with input speed of 7400 rpm. Output speed shall remain within the limits specified in 3.8.4.1.

4.3.2.1.4 Combined Input Speed/Load Changes With input speed of 3700 rpm and 0 load, accelerate the input speed to 7400 rpm at a rate of 900-1500 rpm per second. During the acceleration apply a shock load of 100% rated load. With input speed of 7400 rpm and zero load, decelerate the input speed to 3700 rpm at a rate of 900-1500 rpm per second. During the deceleration apply a shock load of 100% rated load. Output speed shall remain within the limits specified in 3.8.4.2.

4.3.2.1.5 Overload Test - With input speed of 3700 rpm and 100% rated load, increase the load to 150% rated load and maintain for 5 minutes. Repeat this test at 7400 rpm. The outspeed speed regulation shall remain within the limits of 3.8.3.1.

4.4.9 Instrumentation Tolerances - A tolerance of ±5°C is allowed for all temperature

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conditions; ±5 percent on pressure measurements; ±50 rpm on input speed; ±10 rpm on output speed; ±5 percent on loads and fluid flow; ±2 percent on input power; ±1 percent on all other measurements. Instrumentation shall be calibrated prior to testing and at intervals not to exceed 30 days during testing.

4.6.2.3 Temperature/Altitude Test - The transmission shall be operated at average rated input speed and at rated load for one hour in an ambient environment of 149°C and 1.68 psia. Temperature of the oil into the reservoir shall be maintained at 121±5°C. After this test the chamber pressure shall be increased to 22.7 psia, and the transmission shall be operated for one hour at rated load with oil and ambient temperature as indicated above. The transmission shall complete these tests without failure and without impairment of subsequent performance.

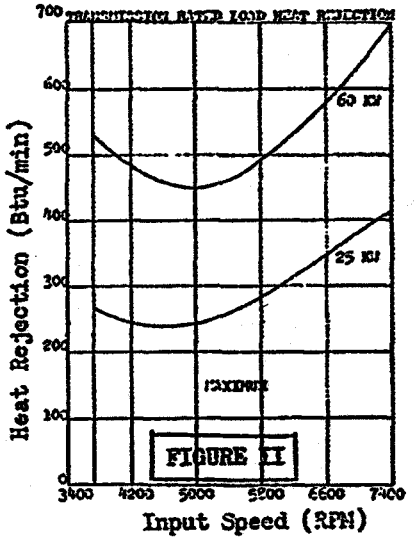


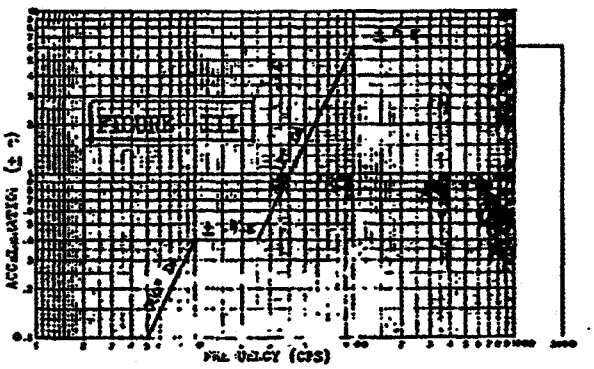
FIGURE II

DELETION OF REQUIREMENTS:

The following requirements of MIL-T-7101 are deleted; 3.3.14, 3.3.18, 3.9.2, 4.6.12, 4.6.14, 4.6.20, and 6.1.

Max. weight (dry)	64 lbs.
Max. overhung moment	355 lb-in
Max. oil flow	11 gal/min
Shear section	8600±600 in-lb
Input speed range	3700-7400 rpm
Rating	60 KW
Output speed	8000 rpm
Oil pressure drop in external circuit	65±35 psi

TABLE 1



LOAD CURVE FOR VIBRATION TEST

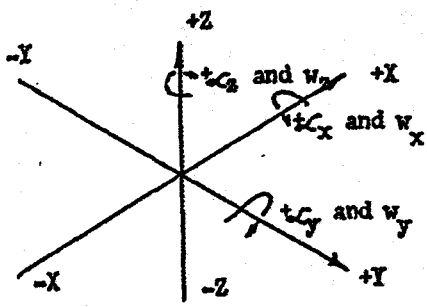


FIGURE IV

	RUN	OIL-IN TEMP. °C	LOAD KVA	PF	RPM	TIME
1.	Check Run (30 Min. Total)	Room - 66 Room - 66 Room - 66	0, 30, 60 0, 30, 60 0, 30, 60	.75 .75 .75	3700 7000 7400	10 Min. 10 Min. 10 Min.
2.	Shut Down	1/2 Hr. Minimum				
3.	Normal Operation (4.5 Hrs. Total)	Room - 37.8 37.8 - 93 93 - 121 121 121 - 93	0 30 30 60 30	--- .75 .75 .95 .75	3700 7400 7000 7000 3700	10 Min. 10 Min. 3 Hrs. 1 Hr. 10 Min.
4.	Shut Down	1/2 Hr. Minimum				
5.	Repeat (1) and (2)					
6.	High Temperature Run (2.5 Hrs. (Total)	Room - 37.8 37.8 - 93 93 - 149 149 149 - 121	0 30 30 60 30	--- .75 .75 .95 .75	3700 7400 7000 7000 3700	10 Min. 10 Min. 1.5 Hrs 0.5 Hrs 10 Min.
7.	Shut Down	1/2 Hr. Minimum				

TABLE 2

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The Quick Attach-Detach Kit described in Figure V is not an integral part of the transmission and is shown for reference and compatibility purposes only. In order to meet the requirements of this Military Specification Sheet it is necessary that all Qualification Tests and all Acceptance Tests be conducted with the transmission mounted to a QAD Kit conforming to Figure V. Adapter plate screws will be attached to the engine and torqued to 28-30 ft-lbs. The bolt on the QAD ring shall be torqued to 12-15 ft-lbs.

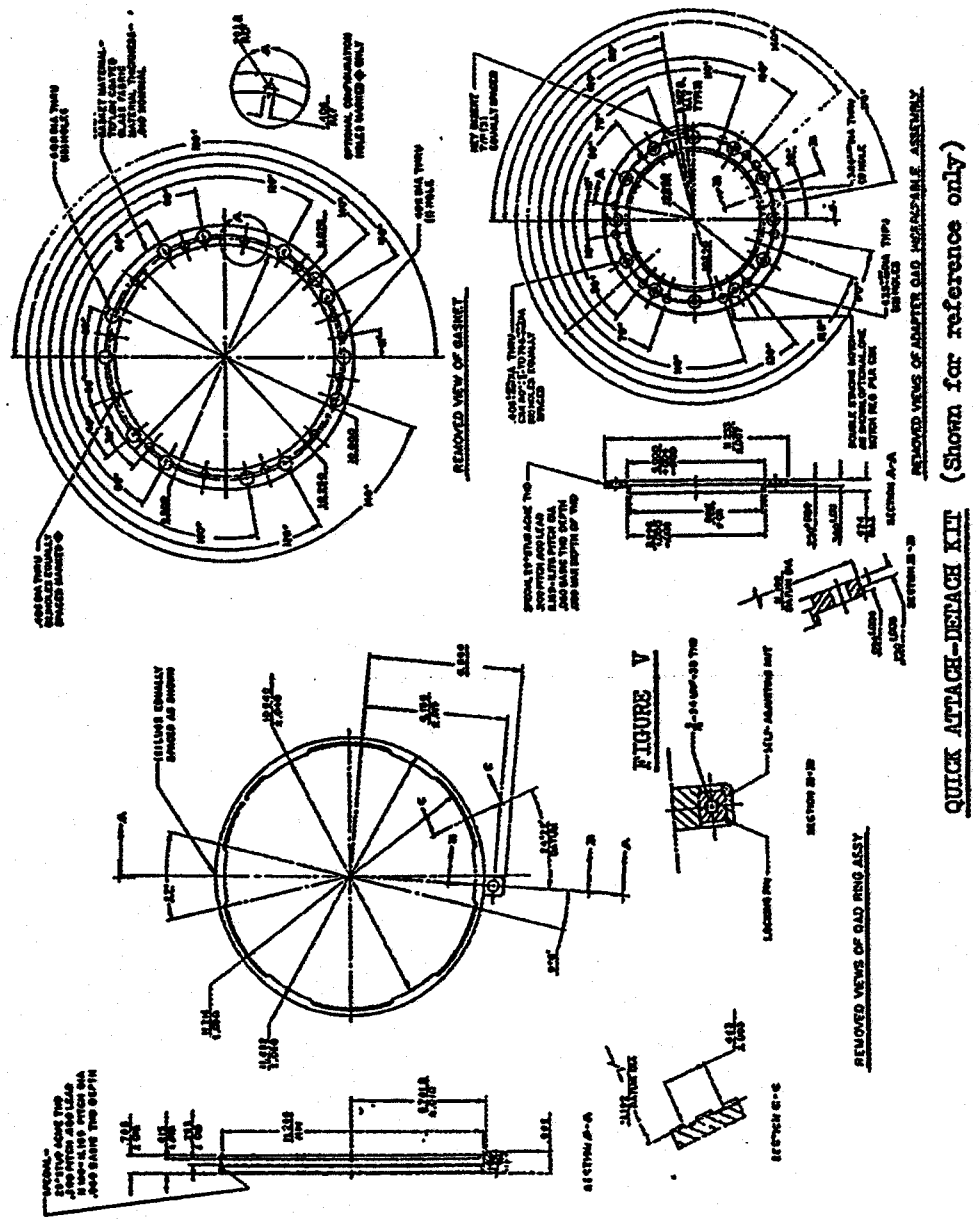


FIGURE V

QUICK ATTACH-DETACH KIT (Shown for reference only)

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